

Counterweight Set - Technical Datasheet

Generic reference template - complete with site-specific details before use

Document Control

Document title	Counterweight Set - Technical Datasheet
Equipment type	Counterweight set / manually operated flying system
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1. Purpose and Scope

This datasheet provides a generic technical reference for counterweight flying sets used in theatre and entertainment environments. It is intended to support equipment identification, inspection planning, maintenance review and portal guidance. It must be completed with site-specific design information, manufacturer documentation and competent-person assessment before being used as a formal record.

2. Equipment Description

- A counterweight set is a manually operated flying system used to raise, lower, support or balance flown bars, scenery, curtains, masking, lighting equipment or other approved suspended loads.
- The system normally relies on a balance between the suspended load and counterweights contained within an arbor or cradle.
- The complete load path may include the flying bar, wire ropes, terminations, loft blocks, head block, arbor, guide rails, hand line / operating line, tensioning arrangements, locks and supporting structure.
- The system should be treated as a complete installed lifting arrangement rather than a collection of unrelated components.

3. Typical Components

Flying bar / batten	Suspended bar used to carry the load. May support scenery, curtains, masking or equipment.
Lift lines / wire ropes	Lines connecting the flying bar to the arbor or counterweight arrangement.
Head block and loft blocks	Sheaves that guide wire ropes and transfer load through the structure.
Counterweight arbor / cradle	Carries counterweights used to balance the suspended load.
Counterweights	Removable weights used to balance the flown load.
Operating line / hand line	Line used by the operator to control manual movement.
Guides / guide rails	Control arbor movement and reduce uncontrolled lateral

	movement.
Locks / brakes / cleats	Devices used to secure or control the set, depending on system design.
Supporting structure	Structural steelwork, grids, beams, brackets, anchors or fixings that support the set.

4. Technical Data to Complete

Asset number / ID	To be completed
Location / venue / area	To be completed
Manufacturer / installer	To be completed
Year installed / commissioned	To be completed
Set number / line number	To be completed
Rated capacity / SWL / WLL	To be completed and verified
Flying bar length and type	To be completed
Number and diameter of lift lines	To be completed
Rope / wire rope specification	To be completed
Head block / loft block details	To be completed
Counterweight type and mass per weight	To be completed
Travel distance	To be completed
Control method	Manual operation / hand line / other - specify
Inspection interval	To be confirmed by competent person or written scheme
Associated documentation	Drawings, certificates, reports, manuals and inspection records

5. Intended Use and Limitations

- Use only for the intended theatre flying or suspension application and within the verified rating of the complete system.
- Do not overload the set by load weight, poor distribution, excessive point loading or unapproved attachments.
- Only trained and authorised persons should load, balance, operate, alter or inspect the set.
- Counterweights should be added and removed in accordance with site procedure to avoid unbalanced or runaway conditions.
- The system should not be modified, re-roped, re-routed or repurposed without competent assessment and documented approval.
- Do not use the system where rating, condition, operation, balance or supporting structure is uncertain.

6. Compliance and Good Practice Context

This section is a summary guide only. Formal dutyholder decisions should refer to the applicable regulations, approved codes of practice, manufacturer information, site procedures and competent-person judgement.

LOLER	Relevant where the counterweight set is lifting equipment or part of a lifting operation. Consider strength, stability, marking, planning, supervision, thorough examination and reporting.
PUWER	Relevant because the counterweight set is work equipment. Consider suitability, maintenance, inspection, safe use, information, instruction and training.
BS EN 17206	Relevant where the set forms part of entertainment machinery or stage machinery arrangements.
ABTT Technical Standards	Industry guidance / good practice for places of entertainment, supporting competent design, installation, inspection, maintenance and operation.

7. Inspection Guidance - What to Check

- Confirm the set is correctly identified and that SWL / WLL / rated capacity information is available and suitable for the intended load.
- Check the flying bar / batten for bending, dents, corrosion, cracked welds, damaged fixings, unauthorised holes or signs of overload.
- Inspect lift lines and wire ropes for broken wires, kinks, bird-caging, corrosion, flattening, wear, abrasion, poor reeving or incorrect termination.
- Inspect rope terminations, clamps, sockets, thimbles, shackles and connection points for correct fitting, tightness, damage and compatibility.
- Check head blocks, loft blocks and sheaves for wear, free rotation, alignment, secure fixing and damaged grooves or bearings.
- Inspect the arbor / cradle for distortion, loose parts, damaged guides, missing retaining components, incorrect weight stacking or signs of impact.
- Check guide rails, tracks or guide systems for alignment, secure fixing, corrosion, obstruction or damage.
- Check the operating line / hand line for wear, cuts, contamination, glazing, poor tension, incorrect routing or insecure termination.
- Check locks, brakes, cleats or securing devices for effective function, correct engagement and visible damage.
- Operate the set where safe to do so and check for smooth travel, abnormal noise, rubbing, snatching, drift, imbalance or uncontrolled movement.
- Confirm that loading, balancing and operating procedures are being followed and that unauthorised users cannot access or alter the system.
- Review inspection records, maintenance history, defect reports, modifications and previous thorough examination reports.

8. Common Defects and Red Flags

- Missing or unclear SWL / WLL / set identification.
- Damaged, corroded, kinked, bird-caged, flattened or incorrectly terminated wire ropes.
- Loose, missing or incorrectly fitted clamps, fixings, bolts, shackles or terminations.
- Worn sheave grooves, seized bearings, misaligned pulleys or damaged blocks.
- Bent or damaged flying bar, arbor, guides, support brackets or structural connections.
- Counterweights not secured, incorrectly stacked or unsuitable for the arbor.
- Set difficult to operate, drifting, snatching, binding, rubbing or making abnormal noise.
- Evidence of unauthorised alteration, additional holes, non-standard repairs or changed load use.
- Missing records, unknown rating, uncertain installation details or unresolved defects.

9. Maintenance and Record Keeping

- Maintain the system in accordance with manufacturer information, competent-person recommendations and site procedures.
- Keep records of inspection, thorough examination, maintenance, defects, repairs, modifications and return-to-service decisions.
- Do not return defective equipment to service until defects have been assessed, corrected and recorded by a competent person where required.
- Review recurring defects to identify root causes such as poor operation, unsuitable loading, inadequate training or insufficient inspection frequency.
- Ensure operators receive appropriate information, instruction and training for the specific system and site procedures.

10. Operational Safety Notes

- Before use, confirm the load, travel path, balance condition and landing / stopping arrangements.
- Keep people clear of suspended loads and moving counterweight/arbor areas unless suitable controls are in place.
- Do not leave the set unsecured or in an unsafe state after loading, balancing or operation.
- Use clear communication when operating or loading the system.
- Report unusual behaviour immediately and withdraw the set from use if safety is uncertain.

11. Completion Sign-Off

Completed by	
Role / company	
Date	
Reviewed by competent person	Yes / No / N/A
Notes / limitations	

Important Disclaimer

This datasheet is a generic reference template. It does not replace manufacturer instructions, design drawings, structural verification, a written scheme of examination, a site-specific risk assessment, or the judgement of a competent person. Site-specific values must be completed and verified before formal use.